# WHO WANTS A TRAUTONIUM MUST BUILD ONE

*Meg Travers*
Western Australian Academy of Performing Arts
Edith Cowan University
Perth, Western Australia, Australia

*A/Prof Cat Hope*
Western Australian Academy of Performing Arts
Edith Cowan University
Perth, Western Australia, Australia

**ABSTRACT**

The man most associated with the development of the Trautonium, Oskar Sala, stated "Wer ein Trautonium will, muss sich eins bauen" – "Who wants a Trautonium, must build one". With only a limited number of the instruments ever made, today if you wish to learn the instrument or perform the works composed for it, first you must build your own. This paper looks at the beginning of the journey to research and build a modern Trautonium, and the implications for the preservation of non-commercial electronic musical instruments to allow them to remain in active use.

**Keywords:** Trautonium, History of electroacoustic music, Music information retrieval, makers, archiving, preservation, modular synthesizer, musicology.

## 1. INTRODUCTION

This project about the 21st century recreation of an example early electronic musical instrument, utilising modern electronics – the Trautonium. Whilst the scores and recordings of the instrument are an essential part of assessing the success of this recreation, they are artefacts whose preservation is well understood and undertaken. The instruments themselves, remain under preservation and on display in various museums, however this circumstance does not lend itself to ongoing performances on the instruments, nor in the future development of it.

## 2. PRESERVATION

Preserving the ability to re-perform works, and not simply the output of performance (such as scores and recordings), is the major part of the archiving, curation and preservation of the performing arts (Giaretta 2011). This is a challenge in the re-performance of works for the Trautonium.

For the composers of the 20th century, the issues of preservation are compounded by their temporal location prior to the digital revolution. These musicians used analogue and early digital electronic devices as part of their sonic palette, and the archiving of these objects is more akin to museology than musicology. Much of the work already done on the preservation of 20th century eletronic music, such as the MUSTICA project (Bachimont, Blanchette et al. 2003), is principally related to the preservation of software, rather than hardware sound sources, digital data rather than analogue.

Since the death of Sala, the Trautonium exists as a small number of bespoke instruments in museums in Germany, exhibited as specimens in the museum. Musicologists, music creators and performers are effectively prohibited from using the instrument as it was intended – for the performance of electronic music. Whilst museums are traditionally seen as places that "acquire, conserve, research, communicate and exhibit the tangible and intangible heritage of humanity" (International Council of Museums), they are generally principally interested in the conservation of the item, and not in its continued usability. The Deutsches Museum website speaks of the need to document and catalogue the material left behind by Sala, and only undertake preservation activities on his recordings and the musical instruments (Deutsche Museum a).

These preservation activities have been undertaken by German engineers with some interest in electronic music, and whilst they have not improved access to the original instrument, they have studied the technology used in the instrument and created variations to bring the instrument into the 21st century. Oskar Sala knew his instrument so well that the controls of the sound generation section had no labels on them to guide a player in how the instruments sounds are created, or what circuits Sala is tweaking in his performances. It is only through the investigative work of these engineers that there is now an understanding of how the instrument worked, and how it can be rebuilt.

## 3. HISTORY

The Trautonium was first developed in 1929 in Germany by Friederich Trautwein and Oskar Sala. Sala, who was the instruments principal performer, passed away in 2002 without having trained anyone in performing on the instrument, and leaving behind seventy shelf meters of notes and recordings of pieces composed for the Trautonium and the instruments themselves in various museums in Germany. However for all intents and purposes, the opportunity for people to perform these works on the instrument they were written for has been closed off. Whilst many commercial units,

electronics kits and software instruments exist to replicate other early electronic instruments such as the Theremin, the work that has been conducted to replicate the Trautonium has been highly specialised and almost entirely centred on museum display, yet the Trautonium may have as large a repertoire as the Theremin available to it.

The Trautonium gave Germany an early entry to the development of electronic musical instruments, following immediately after the development of the Theremin in Russia in 1927, and the Ondes Martenot in France in 1928.

The Deutsche Museum details the variations of the instrument were produced for or by Oskar Sala (Deutsche Museum b).

- The first RVS Trautonium from 1929 is located at the Deutsche Museum in Munich;
- The Rundfunk-Trautonium developed for a German radio station in 1935 has decayed or been destroyed;
- The Konzert-Trautonium developed in 1938 was the first instrument used by Sala for concerts, and is currently located at the Deutsche Museum in Munich;
- The Mixtur-Trautonium was developed by Sala after World War 2 which utilised a new form of sound synthesis previously only theorised, subharmonics, for which he received patents (Sala 1956) (since expired) in the USA, Germany and France. This instrument is located at the Deutsche Museum in Bonn;
- The Mixtur-Trautonium nach Oskar Sala (also known as the Halbleiter Trautonium) was developed for Sala by academic staff and students from the University of applied sciences of the Deutsche Bundespost Berlin from 1980-1988. This variation utilised microelectronics and semiconductors. This instrument was the principle version Sala used until his death in 2002. It is currently stored with the Berlin Musical Instrument Museum.

Additionally, between 1933 and 1935, Telefunken manufactured less than 100 “Volkstrautoniums” based on the original version of the Trautonium, but the high price (roughly two and a half months of the average salary) did not encourage people to purchase the instrument and it was discontinued after only thirteen were sold (120years.net) and of those, less than ten are believed to still exist, three of which are with the Deutsche Museum in Munich.

Unlike the Theremin and Ondes Martenot, the Trautonium is rarely used in current music practice, either in art music, popular music or film soundtracks for which it became principally known.

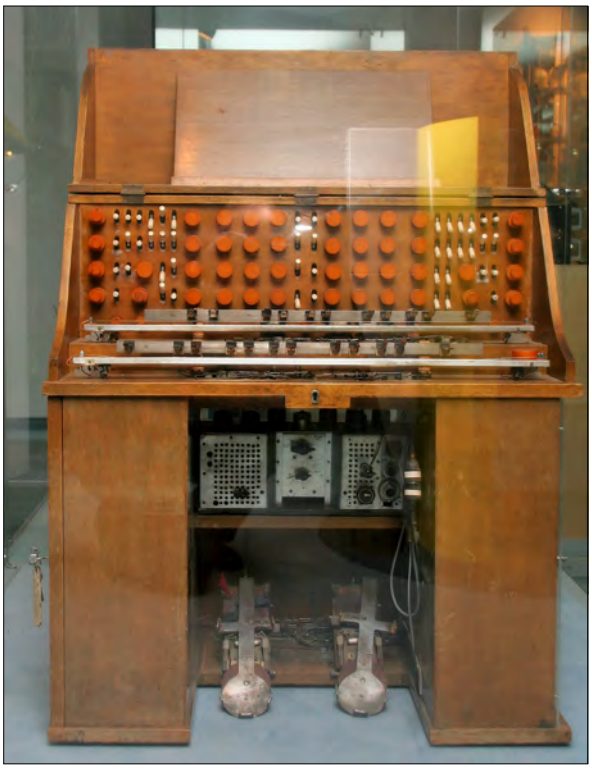


**Figure 1**. Mixtur-Trautonium on display in Deutsche Museum, in Bonn (photo by MatthiasKabel, Wikipedia). Note the lack of labeling on the controls.

## 4. LATER VERSIONS

The Trautonium has not been completely ignored by people wishing to recreate the instrument, and a small number of versions have already been developed.

### 4.1. Replica Mixtur-Trautonium (Vienna)

Doepfer MusikElektronik were contracted by the Technishes Museum in Vienna to construct the sound generation electronics of a replica Mixtur-Trautonium for display in the museum. Dieter Doepfer was able to consult with Oskar Sala in the development of this instrument, and this led to his company creating several synthesizer modules to replicate the sounds of the Mixtur-Trautonium, the prototype modules being installed in that instrument. (Doepfer MusikElectronic GMBH a).

### 4.2. Trautoniks

Trautoniks are a German company who have the highly specialised business of building faithful reproductions of the Volkstrautonium. Their initial work with the Trautonium was in the development of the electronics for the Mixtur-Trautonium nach Oskar Sala, and providing repairs to this instrument for Oskar Sala in 1990-1995 (Trautoniks 2012). Since then they have built one clone of this instrument for the Berlin Government, and three Volkstrautoniums for Peter Pichler who has recently started performing on these instruments in Germany.

Sala’s involvement with these later projects as well as his use of the Mixtur-Trautonium nach Oskar Sala,

demonstrations that he was interested in how the Trautonium may develop using advances in technology. However, given all of these changes to the instrument over time raises the question of what is quintessentially, the Trautonium?

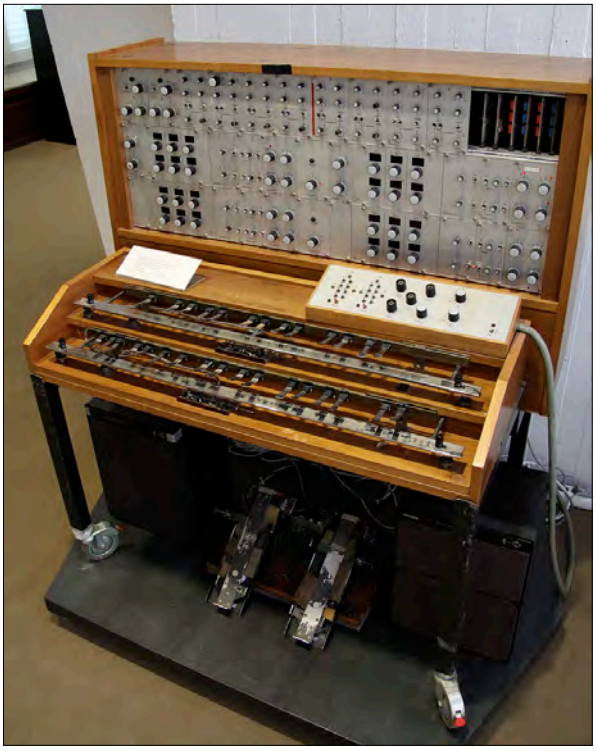


**Figure 2**. The Mixtur-Trautonium nach Oskar Sala, a much modernized version of the instrument created between 1980-1988.

## 5. SYNTHESIS AND SOUNDS

Unlike the Theremin and the Ondes Martenot which produced sine waves through heterodyning, the Trautonium generated a sawtooth wave that could be passed through a set of filters (a first example of what is now known as Subtractive Synthesis) and demonstrated Trautwein's theories which were that a sound is not so much characterized by its fundamental frequency, as by the frequency and dampening of the secondary vibrations setup by it which he referred to as formants (Kock 2012).

Apart from this new process for synthesising interesting sounds, the method of playing the Trautonium was also unique, with the performer pressing a wire onto a metal plate (known as the "manual") - an action that is somewhere between playing keyboard and stringed instruments - thus changing the resistance of the circuit and altering the frequency of the sound produced. This method allowed not only the glissandi type of playing that the Theremin and Ondes Martenot produced, but also the ability to play individual notes quite percussively and precisely.

As well as developing subharmonic synthesis in the Mixtur-Trautonium, in 1965, Sala commissioned the Berlin based measuring instruments company, Kamphausen GMBH, to develop for him a Frequency Shifting device based on tube technology. This unit was a one-off, bespoke device and became Sala's "musical calling card" (Trautonist 2012). At this time, devices of this type were rarely seen in the field of music, coming to prominence in the 1970's when commercial units were developed specifically for musical use by Bode and Moog. Developed initially to combat feedback in communication and amplification systems, Frequency Shifters work by moving all audio frequencies by a specific frequency amount (for example all are moved by 100Hz). By comparison, Pitch Shifters work in a more "musical" way by moving the frequencies by intervals (eg: all frequencies are doubled). When using a Frequency Shifter, the overtones in the source sound will become increasingly distorted as the shift is increased, and it is this sound that Sala made extensive use of in his performance and compositions.

## 6. CHOOSING A NEW TRAUTONIUM

Even in an isolated location such as Western Australia, choosing to learn a musical instrument usually begins with selecting one from a range available at your store or school. Choosing to learn the Trautonium begins with research, then deciding how to build or procure one from Germany. One important decision was made early regarding the naming of the instrument to be built, and so in the Australian idiom, it has become known as "The Traut".

### 6.1. Sounds and Circuits

The instruments built by Trautoniks are considered to be the most faithful reproductions. The low end Volkstrautoniums have a price of 5,499 Euros (around $8,000 AUD), which similarly to the instrument it is reproducing, is some months of the average wage now.

Importantly, the Volkstrautoniums did not include the subharmonic synthesis that Sala developed later, and so many of the later works created for the Mixtur-Trautonium may be unplayable on this model.



**Figure 3**. A Volkstrautonium, a far simpler instrument than the Mixtur-Trautonium.

Dieter Doepfer is an engineer and designer of modular synthesizers, and was able to work with Oskar Sala prior to his death, in designing modular synthesizer

components, which could replicate the functioning of the Mixtur-Trautonium for the replica instrument located in the museum in Vienna. Doepfer are a widely known and used brand of modular synthesizer who have been designing synthesizers since 1977 and pioneered the Eurorack format. Obtaining their modules is comparatively straightforward and affordable.

The three specific modules which they designed to work with others in their range are the A-104 Trautonium Formant Filter, the A-113 Subharmonic Generator and the A-198 Trautonium Manual, and they have suggested how these can be incorporated with their other analogue modules to create a modular Trautonium (Doepfer Musikelektronik GMBH b). The modules range in price, but a complete system can be purchased as modules for around $2,500 AUD.

One notable issue with the Doepfer system is that of a Frequency Shifter. Whilst Doepfer previously made the A-126 Eurorack Frequency Shifter, this has been out of production for some time due to the unavailability of a specific integrated circuit (the CEM3382) which was an essential part of their Frequency Shifter circuit design. This is another example of the variety of issues in the preservation or recreation of old electronic devices.

Whilst both the Trautoniks and Doepfer systems are both attempts to recreate the Trautonium, their differences are multiple. Firstly the Trautoniks instrument is based on the far simpler Volkstrautonium design, including its industrial design which is faithful to the 1930's instrument. The Doepfer instrument has been created as sound synthesizer, not to look historically correct, but to be part of a standard modular instrument, such as is used and toured by musicians around the world even today.

## 6.2. Robustness and Future Proofing

Whilst an instrument that can sound and adequately be performed as a Trautonium is the principle desired outcome, building an instrument which is able to withstand being transported to performances, and also be connected to other electronic instruments are reasonable features for a musical instrument in the 21st century.

The Trautoniks instrument though having a vintage visual charm, is built into a wooden case that is an intrinsic part of the instrument, and damage to this case is equally damage to the instrument. It is a monolithic system that does not include any MIDI implementation.

Doepfer's modular system can be built into many housings, and they have available modules to connect to other systems via both MIDI and Control Voltage.

These issues along with the critical presence of a Subharmonic Generator make the case for building the instrument from Doepfer modules, with the exception of the Frequency Shifter which is another Eurorack module – Synthesis Technology E560 Deflector Shield.

# 7. THE TRAUT

## 7.1. Building

Once the Doepfer modules arrived, a suitable housing was required to build them into. Discussion with another German company Clicks and Clocks followed, and they were able to build a custom set of Eurorack rails which can be installed as one piece into a standard 6RU roadcase.

Doepfer were also able to provide a kit form power supply and buss boards that were installed into the rear of the case. Germany's standard household electrical voltage is very close to Australia's, and all that was needed was an adapter to change from the European plug format to the Australian one.

The assembled instrument does not look anything like the Mixtur-Trautonium, but does bear some resemblance to the controls on the Mixtur-Trautonium nach Oskar Sala, having a shared aluminium faceplate aesthetic. Importantly, this new version is also much smaller than the original and rebuilt Mixtur-Trautoniums, fitting neatly into the 6RU case with the exception of the Manual.

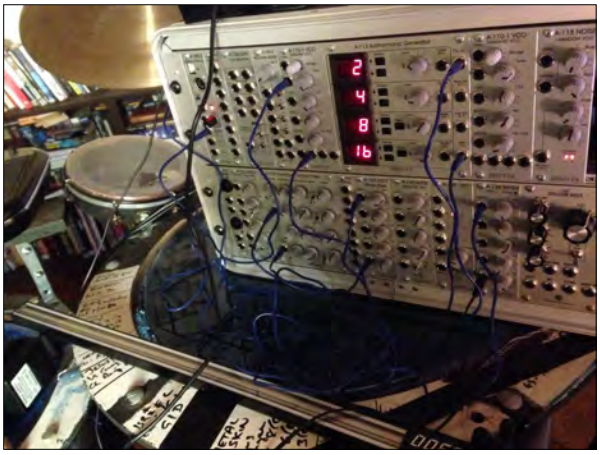



**Figure 2**. Rebuilt Trautonium from Doepfer modules.

## 7.2. Archives and learning to play

Re-performing a late 20th century computer music piece for specific electronic instrumentation would require access to readable synthesizer 'patches' as well as the instrument or software to run these on to recreate the correct sounds. Whilst these are known challenges in the practise of digital archiving even of contemporary music, unfortunately this kind of documentation is not generally available for analogue electronic instruments, and even if there were, it is unlikely that files from the 1970's and earlier would be readable now.

Whilst the instrument is now built, as noted Sala did not document what each of the controls on his instrument were, and so a process of listening to the functions of each module on this new instrument is now being followed, along with close listening and viewing of Sala performing, attempting to connect sounds with

specific areas of control on the instrument, and translate those into the new system.

Sala left his instrument and seventy shelf meters of archives to the Deutsche Museum, including a large number of tapes of “sound studies” which document his development of many of the classic sounds of the Trautonium, including its most famous performance in Alfred Hitchcock’s *The Birds*.

It is hoped that a combination of research into his notes as well as close listening to recordings of performances and sketches in his archive, will provide a way to recreate the sounds that Sala used in the performance of music for the Trautonium. It is an unusual way to approach learning to play a musical instrument, but though the Trautonium is less than 100 years old, it is the only method available.

## 8. CONCLUSION

It has been suggested that most electronic and computer based musical instruments are unstable and ephemeral which does not encourage performers to become virtuosos in their playing (Freed, Uitti et al. 2013). Whilst this was intended as a comment on digital instruments, the same is true for analogue electronic instruments.

Tension lies between the need for stable and available instruments for performers to make use of, and the current situation where many of these older electronic instruments are, if accessible at all, on display in museums where the focus is on preserving the machine rather than making it available for use as a performance instrument. For these reasons, re-creating a low cost and robust Trautonium becomes desirable to allow the instrument to be learned by others in the future, and to enable its future development as technology progresses.

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